Johan Sando

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Experience

Undergraduate Researcher HERC Lab, TX
J. Mike Walker '66 Department of Mechanical Engineering August 2026 - Present

- Developing a high level control strategy framework in Python to estimate gait phase from a dual IMU system (thigh and shank) using a modified Extended Kalman Filter (EKF)
- Implementing and comparing direct user torque inputs and generated torque profiles versus autonomous biomimetic torque profiles, using t-tests and F-tests to analyze variance between control strategies to make a statistically significant claim
- Authoring an undergraduate thesis for Texas A&M's Undergraduate Research Scholars Thesis Program regarding user preference between autonomous and direct control strategies utilizing Extended Kalman Filter-based gait estimation, with the long-term goal of targeting publication through IEEE robotics venues

Undergraduate Research Assistant HERC Lab, TX
J. Mike Walker '66 Department of Mechanical Engineering September 2025 - February 2026

- Designed and developed a custom piece of testing hardware for a biomechanical research paper, utilizing CAD (SolidWorks) and interacting with additive manufacturing requirements to ensure the end result was a verified and valid product
- Translated biomechanical requirements into functional CAD components, working closely with research paper's grad student, iteratively, to ensure the core concepts were accurately represented in the final product
- Collaborated with the corresponding graduate student on a consistent schedule to ensure that the project was hitting milestones in a timely manner and consistently reviewing budgetary limitations to gauge feasibility of the project's design

STEM Tutor (Math/Physics/Chemistry) Blinn College
Blinn College District Learning Center March 2024 – Present

- Tutored undergraduate students in a variety of STEM subjects ranging from Calculus I–III & Differential Equations to Organic Chemistry, with a focus on breaking down complex topics into a methodical approach that strengthen intuition instead of relying on raw memorization
- Identified conceptual gaps in students' understanding of the material and reinforced the core mechanics behind the problem through sufficient explanation and specific problems that emphasized on the targeted mechanics
- Adapted explanations of the material to the student's mechanical maturity and the use of analogies to build mechanical intuition

Projects

Dynamics Modeling of a Double Cart-Pole System July 2026 - Present

- Derived non-linear equations of motion for a cart-mounted inverted double pendulum using Lagrangian mechanics to model conjoined translational and rotational dynamics
- Formulated a continuous-time state-space representation of the double cart-pole system to better understand the foundation for underactuated balancing mechanics

Motorized Ultrasonic Scanning Platform June 2026 - Present

- Developing a motor-driven rotating platform integrated with a ultrasonic sensor for scanning capabilities
- Working on implementing automated stopping logic when objects are detected within a set range
- Designing and 3D printing a custom housing to properly hold the 3+ component integrated system

Education

Texas A&M University College Station, TX
B.S. in Biomedical Engineering — Major/Overall GPA: 3.85/3.57 Expected May 2027

Blinn College Bryan, TX
Chancellor's Scholar & James Durham Scholarship recipient; 4.0 GPA 2022-2024